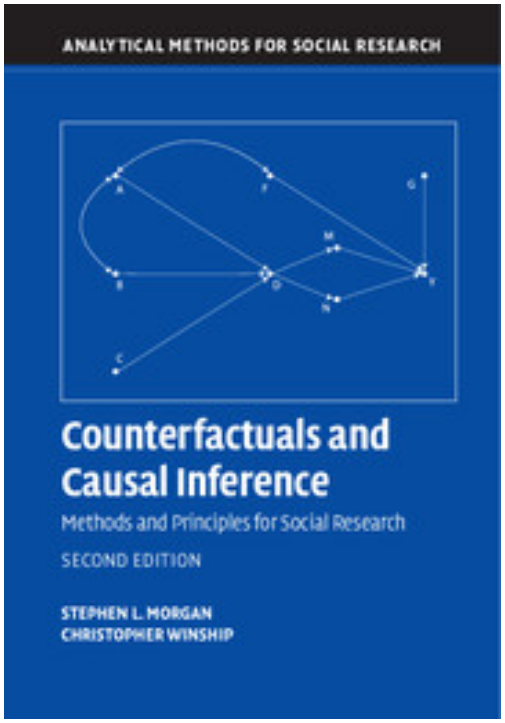
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 6

Outline

1. Regression as a Descriptive Tool
2. Regression Adjustment as a Strategy to Estimate Causal Effects
3. Regression as Conditional-Variance-Weighted Matching
4. Regression as an Implementation of a Perfect Stratification
5. Regression as Supplemental Adjustment When Matching
6. Extensions and Other Perspectives

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A best-fitting linear approximation

- If X is a collection of variables that are thought to be associated with Y in some way, then the conditional expectation function of Y, viewed as a function in X, is denoted $E[Y|X]$.
- Each particular value of the conditional expectation for a specific realization x of X is then denoted $E[Y|X = x]$.
- Least squares regression yields a predicted surface $\hat{Y} = X\hat{\beta}$, where $\hat{\beta}$ is a vector of estimated coefficients from the regression of the realized values y_i on x_i.

- The predicted surface, $X\hat{\beta}$, does not necessarily run through the specific points of the conditional expectation function, even for an infinite sample, because
  1. the conditional expectation function may be a nonlinear function in one or more of the variables in X
  2. a regression model can be fit without parameterizing all nonlinearities in X.
- An estimated regression surface simply represents a best-fitting linear approximation of $E[Y|X]$ under whatever linearity constraints are entailed by the chosen parameterization of the estimated model.

Regression Demonstration 1

Table 6.1 The Joint Probability Distribution and Conditional Population Expectations for Regression Demonstration 1

	Joint probability distribution of S and D	
	Control group: $D = 0$	Treatment group: $D = 1$
$S = 1$	$\Pr[S = 1, D = 0] = .36$	$\Pr[S = 1, D = 1] = .08$
$S = 2$	$\Pr[S = 2, D = 0] = .12$	$\Pr[S = 2, D = 1] = .12$
$S = 3$	$\Pr[S = 3, D = 0] = .12$	$\Pr[S = 3, D = 1] = .2$
	Potential outcomes under the control state	
$S = 1$	$E[Y^0 \mid S = 1, D = 0] = 2$	$E[Y^0 \mid S = 1, D = 1] = 2$
$S = 2$	$E[Y^0 \mid S = 2, D = 0] = 6$	$E[Y^0 \mid S = 2, D = 1] = 6$
$S = 3$	$E[Y^0 \mid S = 3, D = 0] = 10$	$E[Y^0 \mid S = 3, D = 1] = 10$
	Potential outcomes under the treatment state	
$S = 1$	$E[Y^1 \mid S = 1, D = 0] = 4$	$E[Y^1 \mid S = 1, D = 1] = 4$
$S = 2$	$E[Y^1 \mid S = 2, D = 0] = 8$	$E[Y^1 \mid S = 2, D = 1] = 8$
$S = 3$	$E[Y^1 \mid S = 3, D = 0] = 14$	$E[Y^1 \mid S = 3, D = 1] = 14$
	Observed outcomes	
$S = 1$	$E[Y \mid S = 1, D = 0] = 2$	$E[Y \mid S = 1, D = 1] = 4$
$S = 2$	$E[Y \mid S = 2, D = 0] = 6$	$E[Y \mid S = 2, D = 1] = 8$
$S = 3$	$E[Y \mid S = 3, D = 0] = 10$	$E[Y \mid S = 3, D = 1] = 14$

Predicted surfaces from least squares regression

- A regression of Y on D and S that treats D as a dummy variable and S as an interval-scaled variable would yield a predictive surface of

$$\hat{Y} = -2.71 + 2.69(D) + 4.45(S). \tag{6.1}$$

- An alternative regression that treats S as a discrete variable represented in the estimation routine by dummy variables $S2$ and $S3$ would yield a predictive surface of

$$\hat{Y} = 1.86 + 2.75(D) + 3.76(S2) + 8.92(S3). \tag{2}$$

- This model is also a best linear approximation to the six values of the true conditional expectation $E[Y|D, S]$.

The specific estimated values are

$$
\begin{aligned}
D = 0, S = 1 : \ \hat{Y} &= 1.86, \\
D = 0, S = 2 : \ \hat{Y} &= 5.62, \\
D = 0, S = 3 : \ \hat{Y} &= 10.78, \\
D = 1, S = 1 : \ \hat{Y} &= 4.61, \\
D = 1, S = 2 : \ \hat{Y} &= 8.37, \\
D = 1, S = 3 : \ \hat{Y} &= 13.53.
\end{aligned}
$$

In contrast to the model in Equation (6.1), for this model the variable S is given a fully flexible coding. As a result, parameters are fit that uniquely represent all values of S.

Short and long regression

- If one were to add more variables to the predictors, embedding X in a more encompassing set of variables W, then a new set of least squares estimates $\hat{\gamma}$ could be obtained by regressing Y on W. The estimated surface $W\hat{\gamma}$ then represents a best-fitting, linear-in-the-parameters, descriptive fit to a more encompassing conditional expectation function, $E[Y|W]$.
- Whether one then prefers $W\hat{\gamma}$ to $X\hat{\beta}$ as a description of the variation in Y depends on whether one finds it more useful to approximate $E[Y|W]$ than $E[Y|X]$.
- The former regression approximation is often referred to as the long regression, with the latter representing the short regression.

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Regression models and omitted-variable bias

- Suppose that one is interested in estimating the causal effect of a binary variable D on an observed outcome Y. This goal can be motivated as an attempt to obtain a consistent and unbiased estimate of a coefficient δ in a generic bivariate regression equation,

$$Y = \alpha + \delta D + \varepsilon, \tag{6.3}$$

  where α is an intercept and ε is a summary random variable that represents all other causes of Y.
- When Equation (6.3) is used to represent the causal effect of D on Y without any reference to individual-varying potential outcomes, the parameter δ is implicitly cast as an invariant, structural causal effect that applies to all members of the population of interest.

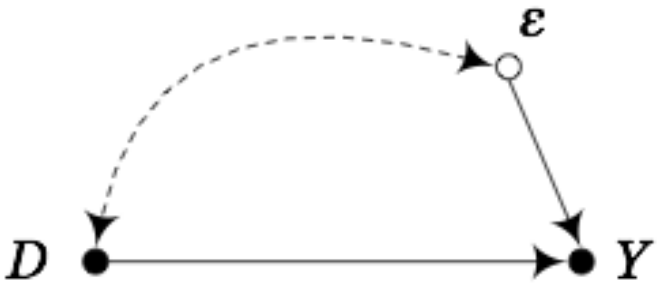
OLS estimators

- The OLS estimator of this bivariate regression coefficient is

$$\hat{\delta}_{\text{OLS, bivariate}} \equiv \frac{\text{Cov}_N(y_i, d_i)}{\text{Var}_N(d_i)}, \tag{6.4}$$

  where $\text{Cov}_N(.)$ and $\text{Var}_N(.)$ are consistent and unbiased, sample-based estimates from a sample of size N of the population-level covariance and variance of the variables.
- Because D is a binary variable, $\hat{\delta}_{\text{OLS, bivariate}}$ is exactly equivalent to the naive estimator, $E_N[y_i|d_i = 1] - E_N[y_i|d_i = 0]$.

(a) A graph in which the causal effect is not identified



(b) An equivalent regression representation of the same graph

Figure 6.1 Graphs for a regression equation of the causal effect of D on Y.

Potential outcomes and bivariate regression

- Recall Equation (4.6), which we reintroduce here as

$$Y = \mu^0 + (\mu^1 - \mu^0)D + \{\upsilon^0 + D(\upsilon^1 - \upsilon^0)\}, \tag{6.5}$$

  where $\mu^0 \equiv E[Y^0]$, $\mu^1 \equiv E[Y^1]$, $\upsilon^0 \equiv Y^0 - E[Y^0]$, and $\upsilon^1 \equiv Y^1 - E[Y^1]$.
- We could rewrite this equation to bring it into closer alignment with the usual representation of bivariate regression by stipulating that $\alpha = \mu^0$, $\delta = (\mu^1 - \mu^0)$, and $\varepsilon = \upsilon^0 + D(\upsilon^1 - \upsilon^0)$.

- The parameters α and δ in Equation (6.3) are usually not considered to be equal to $E[Y^0]$ or $E[\delta]$ for two reasons:
  1. models are usually asserted in the regression tradition without any reference to underlying causal states tied to potential outcomes
  2. the parameters α and δ are usually implicitly held to be constant structural effects that do not vary over individuals in the population (but ... practitioners routinely ignore this and assume that they are average effects ...)
- Similarly, the error term, ε, in Equation (6.3) is almost never separated into two pieces as a function of the definition of potential outcomes and their relationship to D.

Potential outcomes and omitted-variable bias

- Suppose that we are interested in estimating the average treatment effect (ATE), denoted $(\mu^1 - \mu^0)$ here.
- The causal variable D could be correlated with the error term $\upsilon^0 + D(\upsilon^1 - \upsilon^0)$ because of
  1. A *net baseline difference* where υ^0 is correlated with D, generating a correlation between $\{\upsilon^0 + D(\upsilon^1 - \upsilon^0)\}$ and D
  2. A *net treatment effect difference* where $D(\upsilon^1 - \upsilon^0)$ is correlated with D because the average difference in $\upsilon^1 - \upsilon^0$ varies across those in the treatment group and those in the control group
- In either case, an OLS regression of the realized values of Y on D would yield an inconsistent and biased estimate of $(\mu^1 - \mu^0)$.

Consider these two types of bias

Table 6.2 Examples of the Two Basic Forms of Bias for Least Squares Regression

	Differential baseline bias only						
	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	$v_i^0+d_i(v_i^1-v_i^0)$
In treatment group	20	10	5	5	20	1	5
In control group	10	0	−5	−5	0	0	−5
	Differential treatment effect bias only						
	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	$v_i^0+d_i(v_i^1-v_i^0)$
In treatment group	20	10	2.5	0	20	1	2.5
In control group	15	10	−2.5	0	10	0	0
	Both types of bias						
	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	$v_i^0+d_i(v_i^1-v_i^0)$
In treatment group	25	5	5	−2.5	25	1	5
In control group	15	10	−5	2.5	10	0	2.5

Table 6.3 Two-Person Examples in Which Least Squares Regression Estimates Are Unbiased

	Independence of (Y^1, Y^0) from D						
	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	$v_i^0+d_i(v_i^1-v_i^0)$
In treatment group	20	10	0	0	20	1	0
In control group	20	10	0	0	10	0	0
	Offsetting dependence of $Y^1$and Y^0 on D						
	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	$v_i^0+d_i(v_i^1-v_i^0)$
In treatment group	20	10	5	−5	20	1	5
In control group	10	20	−5	5	20	0	5

What is the error term of a regression equation?

- As you can see from the final column in Tables 6.2 and 6.3, the regression error term, $v^0 + D(v^1 - v^0)$, is equal to v^0 for those in the control group and v^1 for those in the treatment group.
- We can rearrange $v^0 + D(v^1 - v^0)$ as $Dv^1 + (1 - D)v^0$ and then rewrite Equation (6.5) as

$$Y = \mu^0 + (\mu^1 - \mu^0)D + \{Dv^1 + (1 - D)v^0\}. \tag{6.6}$$

- It should now be clear that the error term now appears very much like the definition of Y presented earlier as $DY^1 + (1 - D)Y^0$.

- Few presentations of regression characterize the error term of a bivariate regression in this way.
- The connection is made to the counterfactual tradition by specifying Equation (6.3) as

$$Y = \alpha + \delta D + \varepsilon_{(D)}, \tag{6.7}$$

where the error term $\varepsilon_{(D)}$ is considered to be an entirely different random variable for each value of D.

Multiple regression as adjustment in the service of causal analysis

- The basic strategy behind regression analysis as an adjustment technique to estimate a causal effect is to add a sufficient set of "control variables" to the bivariate regression in Equation (6.3).
- The goal is to break a correlation between the treatment variable D and the error term ε, as in

$$Y = \alpha + \delta D + X\beta + \varepsilon^*, \tag{6.8}$$

where X represents one or more variables, β is a coefficient (or a conformable vector of coefficients if X represents more than one variable), ε^* is a residualized version of the original error term ε.

- For the multiple regression analog to the least squares bivariate regression estimator $\hat{\delta}_{\text{OLS, bivariate}}$ in Equation (6.4), the observed data values d_i and x_i are embedded in an all-encompassing $\mathbf{Q}$ matrix, which is $N \times K$, where N is the number of respondents and K is the number of variables in X plus 2 (one for the constant and one for the treatment variable D).
- The OLS estimator for the parameters in Equation (6.8) is then written in matrix form as

$$\hat{\delta}_{\text{OLS, multiple}} \equiv (\mathbf{Q}'\mathbf{Q})^{-1}\mathbf{Q}'\mathbf{y}, \tag{6.9}$$

where $\mathbf{y}$ is an $N \times 1$ vector for the observed outcomes y_i.

- OLS regression is equivalent to the following three-step regression procedure:
  1. Regress y_i on x_i and calculate $y_i^* = y_i - \hat{y}_i$;
  2. Regress d_i on x_i and calculate $d_i^* = d_i - \hat{d}_i$;
  3. Regress y_i^* on d_i^*.
- The regression coefficient on d_i^* yielded by step 3 is the OLS estimate of δ in Equation (6.8), which is typically declared consistent and unbiased for the true value of δ if the correlation between D and ε^* is assumed to be equal to zero.
- Thus, in this simple example, OLS regression is equivalent to estimating the relationship between residualized versions of Y and D from which their common dependence on other variables in X has been "subtracted out."

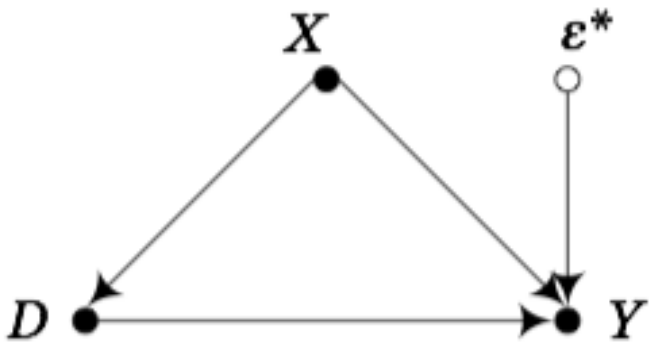


Figure 6.2 A causal graph for a regression equation in which the causal effect of D on Y is identified by conditioning on X.

Now with potential outcomes ...

Table 6.4 Two Six-Person Examples in Which Regression Adjustment Is Differentially Effective

			Regression adjustment with X generates an unbiased estimate for D					
	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	x_i	$v_i^0+d_i(v_i^1-v_i^0)$
In treatment group	20	10	2.5	2.5	20	1	1	2.5
In treatment group	20	10	2.5	2.5	20	1	1	2.5
In treatment group	15	5	−2.5	−2.5	15	1	0	−2.5
In control group	20	10	2.5	2.5	10	0	1	2.5
In control group	15	5	−2.5	−2.5	5	0	0	−2.5
In control group	15	5	−2.5	−2.5	5	0	0	−2.5

			Regression adjustment with X does not generate an unbiased estimate for D					
	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	x_i	$v_i^0+d_i(v_i^1-v_i^0)$
In treatment group	20	10	2.83	2.5	20	1	1	2.83
In treatment group	20	10	2.83	2.5	20	1	1	2.83
In treatment group	15	5	−2.17	−2.5	15	1	0	−2.17
In control group	18	10	.83	2.5	10	0	1	2.5
In control group	15	5	−2.17	−2.5	5	0	0	−2.5
In control group	15	5	−2.17	−2.5	5	0	0	−2.5

Table 6.5 A Rearrangement of the Example in Table 6.4 That Shows How Regression Adjustment Is Differentially Effective

	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	x_i	$v_i^0+d_i(v_i^1-v_i^0)$
				Regression adjustment with X generates an unbiased estimate for D				
				For those with $X=1$				
In treatment group	20	10	2.5	2.5	20	1	1	2.5
In treatment group	20	10	2.5	2.5	20	1	1	2.5
In control group	20	10	2.5	2.5	10	0	1	2.5
				For those with $X=0$				
In treatment group	15	5	−2.5	−2.5	15	1	0	−2.5
In control group	15	5	−2.5	−2.5	5	0	0	−2.5
In control group	15	5	−2.5	−2.5	5	0	0	−2.5
				Regression adjustment with X does not generate an unbiased estimate for D				
	y_i^1	y_i^0	v_i^1	v_i^0	y_i	d_i	x_i	$v_i^0+d_i(v_i^1-v_i^0)$
				For those with $X=1$				
In treatment group	20	10	2.83	2.5	20	1	1	2.83
In treatment group	20	10	2.83	2.5	20	1	1	2.83
In control group	18	10	.83	2.5	10	0	1	2.5
				For those with $X=0$				
In treatment group	15	5	−2.17	−2.5	15	1	0	−2.17
In control group	15	5	−2.17	−2.5	5	0	0	−2.5
In control group	15	5	−2.17	−2.5	5	0	0	−2.5

Now, let's make the connection ...

- Recall that the basic strategy behind regression analysis as an adjustment technique is to estimate

$$Y = \alpha + \delta D + X\beta + \varepsilon^*$$

- The literature on regression often states that an estimated coefficient $\hat{\delta}$ from this regression equation is consistent and unbiased for the average causal effect if ε^* is uncorrelated with D.
- But, because the specific definition of ε^* is conditional on the specification of X, many researchers find this requirement of a zero correlation difficult to interpret and hence difficult to evaluate.

One key special case that shows the connection:

- The crux of the idea, however, can be understood without reference to the error term ε^* but rather with reference to the simpler and more clearly defined error term $\upsilon^0 + D(\upsilon^1 - \upsilon^0)$ or, equivalently, $D\upsilon^1 + (1 - D)\upsilon^0$.
- Regression adjustment by X in will yield a consistent and unbiased estimate of the ATE when
  1. D is mean independent of (and therefore uncorrelated with) $\upsilon^0 + D(\upsilon^1 - \upsilon^0)$ for each subset of respondents identified by distinct values on the variables in X,
  2. the causal effect of D does not vary with X, and
  3. a fully flexible (i.e., dummy-variable with cross-products) parameterization of X is used.

The core of this case:

- Consider the relationship between this set of conditions and what was described earlier as an assumption that treatment assignment is ignorable. Switching notation from S to X results in

$$(Y^0, Y^1) \perp\!\!\!\perp D \mid X, \tag{6.10}$$

  where, again, the symbol $\perp\!\!\!\perp$ denotes independence.
- Rewrite the assumption, deviating Y^0 and Y^1 from their population-level expectations:

$$(v^0, v^1) \perp\!\!\!\perp D \mid X. \tag{6.11}$$

- We need the fully flexible coding of X so that this independence holds within every stratum of the adjustment variables.

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Regression Demonstration 2

- Reconsider Regression Demonstration 1, but now step back from the cautious mindset of the fictitious descriptively oriented researcher.
- Suppose that a causality-oriented researcher had performed the same exercise and obtained the same results for the regression model reported above:

$$\hat{Y} = 1.86 + 2.75(D) + 3.76(S2) + 8.92(S3). \tag{6.12}$$

Under this particular specification of the regression equation, the OLS estimator yields precisely the value of 2.75 in an infinite sample as the sum of sample analogs to three terms:

$$\frac{\text{Var}[D|S=1]\,\Pr[S=1]}{\sum_S \text{Var}[D|S=s]\,\Pr[S=s]}\{E[Y|D=1,S=1]-E[Y|D=0,S=1]\}$$
$$+\frac{\text{Var}[D|S=2]\,\Pr[S=2]}{\sum_S \text{Var}[D|S=s]\,\Pr[S=s]}\{E[Y|D=1,S=2]-E[Y|D=0,S=2]\}$$
$$+\frac{\text{Var}[D|S=3]\,\Pr[S=3]}{\sum_S \text{Var}[D|S=s]\,\Pr[S=s]}\{E[Y|D=1,S=3]-E[Y|D=0,S=3]\} \tag{6.13}$$

First, note that the differences in the braces on the right-hand side of each term are simply the stratum-specific differences in the outcomes, which in this case are

$$E[Y|D=1,S=1]-E[Y|D=0,S=1] = 4-2, \quad (6.14)$$
$$E[Y|D=1,S=2]-E[Y|D=0,S=2] = 8-6, \quad (6.15)$$
$$E[Y|D=1,S=3]-E[Y|D=0,S=3] = 14-10. \quad (6.16)$$

The conditional variances, $\mathrm{Var}[D|S=s]$, contribute to the numerator of each weight as follows:

$$\begin{aligned} &Var[D|S=1]\Pr[S=1] \\ &= \left[\left(\frac{.08}{.08+.36}\right)\left(1-\frac{.08}{.08+.36}\right)\right](.08+.36)\,, \end{aligned} \tag{6.17}$$

$$\begin{aligned} &Var[D|S=2]\Pr[S=2] \\ &= \left[\left(\frac{.12}{.12+.12}\right)\left(1-\frac{.12}{.12+.12}\right)\right](.12+.12)\,, \end{aligned} \tag{6.18}$$

$$\begin{aligned} &Var[D|S=3]\Pr[S=3] \\ &= \left[\left(\frac{.2}{.2+.12}\right)\left(1-\frac{.2}{.2+.12}\right)\right](.2+.12)\,. \end{aligned} \tag{6.19}$$

Table 6.6 The Joint Probability Distribution for Two Variants of the Stratifying and Treatment Variables in Prior Regression Demonstration 1

	Joint probability distribution of S and D	
	Control group: $D = 0$	Treatment group: $D = 1$
	Variant I	
$S = 1$	$\Pr[S = 1, D = 0] = .40$	$\Pr[S = 1, D = 1] = .04$
$S = 2$	$\Pr[S = 2, D = 0] = .20$	$\Pr[S = 2, D = 1] = .04$
$S = 3$	$\Pr[S = 3, D = 0] = .16$	$\Pr[S = 3, D = 1] = .16$
	Variant II	
$S = 1$	$\Pr[S = 1, D = 0] = .40$	$\Pr[S = 1, D = 1] = .04$
$S = 2$	$\Pr[S = 2, D = 0] = .12$	$\Pr[S = 2, D = 1] = .12$
$S = 3$	$\Pr[S = 3, D = 0] = .03$	$\Pr[S = 3, D = 1] = .29$

- For Variant I, the regression model yields

$$\hat{Y} = 1.90 + 3.07(D) + 3.92(S2) + 8.56(S3) \tag{6.20}$$

  for an infinite sample. In this case, the coefficient of 3.07 on D is not particularly close to the unconditional ATE of 2.64, and in fact it is closer to the ATT of 3.33 (although still not particularly close).
- For Variant II, the regression model yields

$$\hat{Y} = 1.96 + 2.44(D) + 3.82(S2) + 9.45(S3). \tag{6.21}$$

For a fully flexible coding of adjustment variables

The OLS estimator $\hat{\delta}_{\text{OLS, multiple}}$ in Equation (6.9) is equal to

$$\frac{1}{c}\sum_{s}\text{Var}_N[d_i|s_i = s]\ \Pr{}_N[s_i = s] \times \tag{6.22}$$

$$\{E_N[y_i|d_i = 1, s_i = s] - E_N[y_i|d_i = 0, s_i = s]\}$$

in a sample of size N. Here, c is a scaling constant equal to the sum (over all combinations of values s of S) of the terms $\text{Var}_N[d_i|s_i = s]\,\Pr_N[s_i = s]$.

Regression Demonstration 3

Table 6.7 The Joint Probability Distribution and Conditional Population Expectations for Regression Demonstration 3

	Joint probability distribution of S and D	
	Control group: $D = 0$	Treatment group: $D = 1$
$S = 1$	$\Pr[S = 1, D = 0] = .4$	$\Pr[S = 1, D = 1] = 0$
$S = 2$	$\Pr[S = 2, D = 0] = .1$	$\Pr[S = 2, D = 1] = .13$
$S = 3$	$\Pr[S = 3, D = 0] = .1$	$\Pr[S = 3, D = 1] = .27$
	Potential outcomes under the control state	
$S = 1$	$E[Y^0 \mid S = 1, D = 0] = 2$	
$S = 2$	$E[Y^0 \mid S = 2, D = 0] = 6$	$E[Y^0 \mid S = 2, D = 1] = 6$
$S = 3$	$E[Y^0 \mid S = 3, D = 0] = 10$	$E[Y^0 \mid S = 3, D = 1] = 10$
	Potential outcomes under the treatment state	
$S = 1$	$E[Y^1 \mid S = 1, D = 0] = 4$	
$S = 2$	$E[Y^1 \mid S = 2, D = 0] = 8$	$E[Y^1 \mid S = 2, D = 1] = 8$
$S = 3$	$E[Y^1 \mid S = 3, D = 0] = 14$	$E[Y^1 \mid S = 3, D = 1] = 14$
	Observed outcomes	
$S = 1$	$E[Y \mid S = 1, D = 0] = 2$	
$S = 2$	$E[Y \mid S = 2, D = 0] = 6$	$E[Y \mid S = 2, D = 1] = 8$
$S = 3$	$E[Y \mid S = 3, D = 0] = 10$	$E[Y \mid S = 3, D = 1] = 14$

- Consider now the estimated values that would be obtained with data arising from this joint distribution for a regression model specified equivalently as in Equations (6.12), (6.20), and (6.21):

$$\hat{Y} = 2.00 + 3.13(D) + 3.36(S2) + 8.64(S3). \tag{6.23}$$

- Imagine that a researcher simply observes $\{y_i, d_i, s_i\}_{i=1}^{N}$ and then estimates the model in Equation (6.23) without first considering the joint distribution of S and D.
- It would be entirely unclear to such a researcher that there are no individuals in the sample (or in the population) whose values for both D and S are 1. Such a researcher might therefore be led to believe that the coefficient estimate for D is a meaningful estimate of the causal effect of D for all members of the population.

Outline

1. Regression as a Descriptive Tool
2. Regression Adjustment as a Strategy to Estimate Causal Effects
3. Regression as Conditional-Variance-Weighted Matching
4. Regression as an Implementation of a Perfect Stratification
5. Regression as Supplemental Adjustment When Matching
6. Extensions and Other Perspectives

Let's not overlook something obvious

- For the data common to these demonstrations, an analyst could specify S as two dummy variables, D as one dummy variable, and include all two-way interactions between S and D.
- In so doing, the analyst has enacted the same perfect stratification of the data by fitting a model that is saturated in both S and D to all of the cells of these examples:

$$\hat{Y} = 2 + 2(D) + 4(S2) + 8(S3) + 0(D \times S2) \; + 2(D \times S3). \quad (6.24)$$

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Table 6.8 Average Bias Comparisons for Selected Matching Estimates of the ATT from Matching Demonstration 4, With and Without Supplemental Regression Adjustment for the Assumed Determinants of Treatment Assignment

	Specification of treatment assignment variables:			
	Incomplete specification		Complete specification	
Method	Unadjusted	Adjusted	Unadjusted	Adjusted
Nearest-neighbor match:				
1 without replacement (MI)	0.75	0.72	0.00	-0.10
1 with replacement (MI)	0.95	0.81	0.19	-0.12
1 with replacement and caliper = .05 SD (MI)	0.98	0.86	0.07	-0.11
5 with replacement (MI)	1.17	0.83	0.50	-0.05
5 with replacement and caliper = .05 SD (MI)	1.19	0.80	0.39	-0.13
Interval match:				
10 fixed blocks (MI)	1.68	0.84	1.68	-0.11
Optimal match (MI-opt)	1.28	0.80	0.54	0.07
Genetic match (MI-gen)	0.96	0.80	0.23	-0.07
Coarsened exact match (cem)	1.28	0.86	0.35	-0.08

Notes: See notes to Table 5.6 for software details.

Why is there usually a reduction in bias?

- Any post-match imbalance that remains in the matching variables is likely to have a component that exists as differences in mean values of the matching variables across the treatment and control cases.
- If we then use a parametric regression model to adjust for the matching variables in the matched datasets, the regression model yields a net estimate of the ATT after a linear adjustment for these lingering mean differences.
- The conditional-variance weighting property of least squares regression estimators remains, but the consequences of this property for estimates is greatly diminished when regression is used in this supplementary way, after the data have already been aligned in pursuit of the ATT.

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Fancier regression

- The implicit weighting scheme of OLS regression is even more complex when linear constraints are placed on many-valued causes.
- For logits and so on, we simply do not have analytic formulas like those we have used to consider what OLS regression is actually doing.
- No one thinks these issues go away for fancier regression because the principles are the same.

The all-cause, complete-specification tradition of regression analysis

- Now that you know how hard it is to estimate a single causal effect with a regression equation, do you think it is generally possible to estimate causal effects for all variables in an estimated regression equation?
- In fact, many additional issues emerge, perhaps most of all p-hacking when using Sherlock Holmes' methods.